

2025/26 Exam Solutions

Section A

Answer BOTH questions from this section.

1. (a) Apply Newton's method

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)},$$

to find the roots of the function

$$f(x) = x^3 + 4x^2 - 10 = 0, \quad (1)$$

using a starting value of $x_0 = 1$. Show your calculations for the first two iterations of the method stating the results to 5 decimal place accuracy. Using MATLAB, Python, Excel or otherwise, continue iterations until two successive solutions agree to 5 decimal places. State the root and the accuracy that has been achieved.

(10 marks)

Solution:

The derivative of $f(x)$ is

$$f'(x) = 3x^2 + 8x,$$

so Newton's method for this equation is

$$x_{k+1} = x_k + \frac{x_k^3 + 4x_k^2 - 10}{3x_k^2 + 8x_k}.$$

2 marks

Using $x_0 = 1$

$$x_1 = \left(1 + \frac{1^3 + 4(1^2) - 10}{3(1^2) + 8(1)}\right) = 1.45455,$$

$$x_2 = 1.45455 + \frac{1.45455^3 + 4(1.45455^2) - 10}{3(1.45455^2) + 8(1.45455)} = 1.36890$$

The iterates are

k	x_k
0	1.00000
1	1.45455
2	1.36890
3	1.36524
4	1.36523
5	1.36523

6 marks

So the root is $x = 1.3652$ correct to 4 decimal places.

2 marks

- (b) Rearrange equation (1) to give an expression for $g(x_k)$ such that

$$x_{k+1} = g(x_k).$$

Use the convergence criteria to show whether this iterative scheme will converge to the root calculated in part (b).

(5 marks)

Solution: Rearranging $f(x)$

$$0 = x^3 + 4x^2 - 10$$

$$10 = x(x^2 + 4x)$$

$$\therefore g(x) = \frac{10}{x^2 + 4x}.$$

The convergence condition is $|g'(x)| \leq 1$. The derivative of $g(x)$ is

$$g'(x) = -\frac{10(x^2 + 4x)}{(x^2 + 4x)^2}.$$

3 marks

Using the root obtained from part (a), $g'(x) = 1.36523 > 1$ so the rearrangement method will not converge for this equation.

2 marks

- (c) The bisection method for computing the roots of a polynomial $f(x)$ begins with two points a and b such that $f(a)f(b) < 0$ and iterates for the following steps:

- Compute the midpoint $c = \frac{1}{2}(a + b)$.
- If $f(c)$ is less than some convergence tolerance then c is the root and the method terminates.
- If $f(a)f(c) < 0$ set $b = c$
- Else set $a = c$

Use the bisection method with starting values of $a = 1$ and $b = 2$ to compute the root of the equation (1). State results to 4 decimal place accuracy and cease iterations when $|f(c)| < 0.5 \times 10^{-4}$. State the value of the root and the accuracy obtained.

(10 marks)

Solution:

k	a	b	c	$f(c)$
0	1.0000	2.0000	1.5000	2.3750
1	1.0000	1.5000	1.2500	-1.7969
2	1.2500	1.5000	1.3750	0.1621
3	1.2500	1.3750	1.3125	-0.8484
4	1.3125	1.3750	1.3438	-0.3510
5	1.3438	1.3750	1.3594	-0.0964
6	1.3594	1.3750	1.3672	0.0324
7	1.3594	1.3672	1.3633	-0.0321
8	1.3633	1.3672	1.3652	0.0001

8 marks

So the root is $x = 1.4$ correct to one decimal place.

2 marks

2. (a) The fourth-order Runge-Kutta method for solving the ODE $y' = f(x, y)$ where $y_0 = y(x_0)$ is

$$\begin{aligned} y_{n+1} &= y_j + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4), \\ k_1 &= f(x_j, y_j), \\ k_2 &= f(x_j + \frac{1}{2}h, y_j + \frac{1}{2}hk_1), \\ k_3 &= f(x_j + \frac{1}{2}h, y_j + \frac{1}{2}hk_2), \\ k_4 &= f(x_j + h, y_j + hk_3). \end{aligned}$$

Apply this method to solve

$$y' = t - y, \quad y(0) = 1$$

at the point where $x = 0.2$ using $h = 0.2$. State your answers to 4 decimal places.

(10 marks)

Solution:

$$\begin{aligned} k_1 &= 0 - 1 = -1, \\ k_2 &= (0 + 0.5(0.2)) - (1 + 0.5(0.2)(-1)) = -0.8, \\ k_3 &= (0 + 0.5(0.2)) - (1 + 0.5(0.2)(-0.8)) = -0.82, \\ k_4 &= (0 + 0.2) - (1 + 0.2(-0.82)) = -0.636, \\ y_1 &= 1 + \frac{0.2}{6}(-1 + 2(-0.8) + 2(-0.82) - 0.636) = 0.8375, \\ k_1 &= 0.2 - 0.8365 = -0.6375, \\ k_2 &= (0.2 + 0.5(0.2)) - (0.8375 + 0.5(0.2)(-0.6375)) = -0.4737, \\ k_3 &= (0.2 + 0.5(0.2)) - (0.8375 + 0.5(0.2)(-0.4737)) = -0.4901, \\ k_4 &= (0.2 + 0.2) - (0.8375 + 0.2(-0.4901)) = -0.3394, \\ y_2 &= 0.8375 + \frac{0.2}{6}(-0.6375 + (2 - 0.4737) + 2(-0.4901) - 0.3394) = 0.7460. \end{aligned}$$

Therefore, $y(0.4) = 0.7460$.

- (b) The first-order Euler and second-order Adams-Bashforth methods for solving the ODE $y' = f(x, y)$ where $y_0 = y(x_0)$ are

$$\text{Euler :} \quad y_{j+1} = y_j + hf(x_j, y_j),$$

$$\text{Adams-Bashforth :} \quad y_{j+1} = y_j + \frac{h}{2}(3f(x_j, y_j) - f(x_{j-1}, y_{j-1})).$$

Using the Euler method for the first step, and the Adams-Bashforth method for all other steps, solve the ODE from part (a) to $x = 1$ using $h = 0.2$.

(10 marks)

Solution:

Euler's method:

$$y_1 = 1 + 0.2(0 - 1) = 0.8,$$

$$t_1 = 0 + 0.2 = 0.2,$$

$$y_2 = 0.8 + \frac{0.2}{2}(3(0.2 - 0.8) - (0 - 1)) = 0.84,$$

$$t_2 = 0.2 + 0.2 = 0.4,$$

$$y_3 = 0.84 + \frac{0.2}{2}(3(0.4 - 0.84) - (0.2 - 0.8)) = 0.856,$$

$$t_3 = 0.4 + 0.2 = 0.6,$$

$$y_4 = 0.856 + \frac{0.2}{2}(3(0.6 - 0.856) - (0.4 - 0.84)) = 0.8744,$$

$$t_4 = 0.6 + 0.2 = 0.8,$$

$$y_5 = 0.8744 + \frac{0.2}{2}(3(0.8 - 0.8744) - (0.6 - 0.856)) = 0.8926.$$

Therefore, $y(1) = 0.8926$.

- (c) Which of the methods used in parts (a) and (b) would you expect to give the more accurate solution to this ODE and why? State ONE way in which the accuracy of the solution to an ODE can be improved.

(5 marks)

Solution: The fourth-order Runge-Kutta would give a more accurate solution than the second-order Adams-Bashforth method since the maximum error is $O(h^4)$ compared to $O(h^2)$.

The accuracy of a numerical solution to an ODE can be improved by using a higher order method or through reducing the step length.

Section B

Answer any TWO questions from this section.

3. (a) Lagrange interpolation is a method for fitting a polynomial to a set of points (x_i, y_i) such that

$$y = L_0(x)y_0 + L_1(x)y_1 + \cdots + L_n(x)y_n,$$

where

$$L_i(x) = \frac{(x - x_0)(x - x_1) \cdots (x - x_n)}{(x_i - x_0)(x_i - x_1) \cdots (x_i - x_n)},$$

with $(x - x_i)$ missing from the numerator and $(x_i - x_i)$ missing from the denominator. Use Lagrange interpolation to find the polynomial which passes through the points $(1, 2)$, $(3, 5)$, $(4, 4)$ (8 marks)

Solution:

$$L_0(x) = \frac{(x - 3)(x - 4)}{(1 - 3)(1 - 4)} = \frac{x^2 - 7x + 12}{6},$$

$$L_1(x) = \frac{(x - 1)(x - 4)}{(3 - 1)(3 - 4)} = \frac{x^2 - 5x + 4}{-2},$$

$$L_2(x) = \frac{(x - 1)(x - 3)}{(4 - 1)(4 - 3)} = \frac{x^2 - 4x + 3}{3}.$$

Therefore,

$$\begin{aligned} y &= 2 \left(\frac{x^2 - 7x + 12}{6} \right) + 5 \left(\frac{x^2 - 5x + 4}{-2} \right) + 4 \left(\frac{x^2 - 4x + 3}{3} \right) \\ &= \frac{x^2 - 7x + 12}{3} - \frac{5x^2 - 25x + 20}{2} + \frac{4x^2 - 16x + 12}{3} \\ &= \frac{2x^2 - 14x + 24 - 15x^2 + 75x - 60 + 8x^2 - 32x + 24}{6} \\ &= \frac{-5x^2 + 29x - 12}{6}. \end{aligned}$$

- (b) An interpolating polynomial passing through the points (x_i, f_i) can be determined by divided differences such that

$$\begin{aligned} y &= f[x_0, x_1] + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \cdots \\ &\quad + f[x_0, x_1, \dots, x_n](x - x_0)(x - x_1) \cdots (x - x_{n-1}) \end{aligned}$$

where the divided differences are defined by

$$\begin{aligned} f[x_i, x_{i+1}] &= \frac{f_i - f_{i-1}}{x_i - x_{i-1}}, \\ f[x_0, x_1, \dots, x_n] &= \frac{f[x_1, x_2, \dots, x_n] - f[x_0, x_1, \dots, x_{n-1}]}{x_n - x_0}. \end{aligned}$$

- (i) Copy down and complete the following divided difference table for the points from part (a) and determine the interpolating polynomial that fits these points.

x_i	f_i	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$
1	2		
3	5	...	...
4	4	...	

(5 marks)

Solution:

x_i	f_i	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$
1	2		
3	5	$\frac{3}{2}$	$-\frac{5}{6}$
4	4	-1	

$$\begin{aligned}
 y &= 2 + \frac{3}{2}(x-1) - \frac{5}{6}(x-1)(x-3) \\
 &= 2 + \frac{3}{2}x - \frac{3}{2} - \frac{5}{6}(x^2 - 4x + 3) \\
 &= -2 + \frac{29}{6}x - \frac{5}{6}x^2.
 \end{aligned}$$

- (ii) An additional point $(8, -12)$ is added to the set of points from part (a). Determine the interpolating polynomial that passes through all FOUR points. (7 marks)

Solution:

x_i	f_i	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$	$f[x_i, x_{i+1}, x_{i+2}, x_{i+3}]$
1	2			
3	5	$\frac{3}{2}$	$-\frac{5}{6}$	
4	4	-1	$-\frac{3}{5}$	$\frac{1}{30}$
8	-12	-4		

$$\begin{aligned}
 y &= -2 + \frac{29}{6}x - \frac{5}{6}x^2 + \frac{1}{30}(x-1)(x-3)(x-4) \\
 &= -2 + \frac{29}{6}x - \frac{5}{6}x^2 + \frac{1}{30}(x-1)(x^2 - 7x + 12) \\
 &= -2 + \frac{29}{6}x - \frac{5}{6}x^2 + \frac{1}{30}(x^3 - 8x^2 + 19x - 12)
 \end{aligned}$$

$$= -\frac{12}{5} + \frac{82}{15}x - \frac{11}{10}x^2 + \frac{1}{30}x^3.$$

- (c) State ONE advantage of determining the interpolation polynomial using divided differences over Lagrange interpolation. State why we couldn't use Newton-Gregory forward differences in this case. (5 marks)

Solution:

Divided differences are more computationally efficient and allow for higher order interpolated polynomials to be determined when given additional points without starting all over again.

3 marks

Newton-Gregory forward differences requires equally spaced x_i values.

2 marks

4. (a) Compute the solution to the following system of linear equations using Gaussian elimination. Round intermediate values to 4 decimal places and state the final solution to 3 decimal places.

$$\begin{pmatrix} 1 & -5 & -1 \\ 7 & -1 & 3 \\ 5 & -3 & 11 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 8 \\ -3 \\ -4 \end{pmatrix}$$

(10 marks)

Solution:

$$\begin{pmatrix} 1 & -5 & -1 & 8 \\ 7 & -1 & 3 & -3 \\ 5 & -3 & 11 & -4 \end{pmatrix} \begin{matrix} R_2 - 7R_1 \\ R_3 - 5R_1 \end{matrix} \rightarrow \begin{pmatrix} 1 & -5 & -1 & 8 \\ 0 & 34 & 10 & -59 \\ 0 & 22 & 16 & -44 \end{pmatrix} \begin{matrix} \\ R_3 - \frac{11}{17}R_2 \end{matrix}$$

$$\rightarrow \begin{pmatrix} 1 & -5 & -1 & 8 \\ 0 & 34 & 10 & -59 \\ 0 & 0 & 9.5294 & -5.8235 \end{pmatrix}$$

4 marks

Solving through back substitution

$$x_3 = \frac{-5.8235}{9.5294} = -0.6111,$$

$$x_2 = \frac{1}{34}(-59 - 10(-0.6111)) = -1.5556,$$

$$x_1 = 8 + 5(-1.5556) + (-0.6111) = -0.3891$$

4 marks

So the solution is $x_1 = -0.389$, $x_2 = -1.556$ and $x_3 = -0.6111$ correct to 3 decimal places. **2 marks**

- (b) Re-cast the matrix equation from part (a) into a diagonally dominant form. Using starting values of $x_i^{(0)} = 0$, compute the first 6 iterations of the Gauss-Seidel method rounding all values to 4 decimal places. State the solutions to the accuracy that has been achieved.

(10 marks)

Solution:

Diagonal dominant form:

$$\begin{pmatrix} 7 & -1 & 3 \\ 1 & -5 & -1 \\ 5 & -3 & 11 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -3 \\ 8 \\ -4 \end{pmatrix}$$

Gauss-Seidel method:

$$\begin{aligned} x_1^{(k+1)} &= \frac{1}{7}(-3 + x_2^{(k)} - 3x_3^{(k)}), \\ x_2^{(k+1)} &= -\frac{1}{5}(8 - x_1^{(k+1)} + x_3^{(k)}), \\ x_3^{(k+1)} &= \frac{1}{11}(-4 - 5x_1^{(k+1)} + 3x_2^{(k+1)}). \end{aligned}$$

Iterations:

k	x_1	x_2	x_3
0	0.0000	0.0000	0.0000
1	-0.4286	-1.6857	-0.6286
2	-0.4000	-1.5543	-0.6057
3	-0.3910	-1.5571	-0.6106
4	-0.3893	-1.5558	-0.6110

So the solution is $x_1 = -0.39$, $x_2 = -1.56$ and $x_3 = -0.61$ correct to 2 decimal places.

- (c) Give ONE advantage and ONE disadvantage of using the Gauss-Seidel method instead of Gaussian elimination to solve a system of linear equations.

(5 marks)

Solution:

The Gauss-Seidel is preferable to Gaussian elimination is more efficient than Gaussian elimination for solving large sparse systems.

2 marks

The Gauss-Seidel method is not guaranteed to converge to a solution for every system.

3 marks

5. (a) The second-order finite-difference approximations to the first and second derivative of $y(x)$ are

$$y'(x) = \frac{y(x+h) - y(x-h)}{2h},$$

$$y''(x) = \frac{y(x+h) - 2y(x) + y(x-h)}{h^2},$$

for a constant step length h . Use these finite-difference approximations to form a system of linear equations that approximate the solution to the following boundary value problem using 4 intervals of equal width

$$y'' + 4y' - 2y = 0, \quad y(0) = 1, \quad y(1) = 2.$$

(10 marks)

Solution:

Let $y_{i\pm 1} = y(x \pm h)$

$$\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} + 4 \left(\frac{y_{i+1} - y_{i-1}}{2h} \right) - 2y_i = 0$$

$$2(y_{i+1} - 2y_i + y_{i-1}) + 4h(y_{i+1} - y_{i-1}) - 4h^2 y_i = 0$$

$$(2 - 4h)y_{i-1} - 4(1 + h^2)y_i + (2 + 4h)y_{i+1} = 0.$$

5 marks

Using 4 intervals then $h = 0.25$ and

$$y_0 = 1,$$

$$y_0 - 4.25y_1 + 3y_2 = 0,$$

$$y_1 - 4.25y_2 + 3y_3 = 0,$$

$$y_2 - 4.25y_3 + 3y_4 = 0,$$

$$y_4 = 2.$$

Since $y_1 = 1$ and $y_2 = 2$ then

$$-4.25y_1 + 3y_2 = -1,$$

$$y_1 - 4.25y_2 + 3y_3 = 0,$$

$$y_2 - 4.25y_3 = -6.$$

5 marks

Students may express the equations in terms of h

- (b) Compute the solution to the boundary value problem by solving the linear system from part (a).

(10 marks)

Solution:

Solving the linear system gives $y_1 = 1.3472$, $y_2 = 1.5751$ and $y_3 = 1.7824$ correct to 4 decimal places.

10 marks

Students may use any method at their disposal to solve the system

- (c) What would you expect to happen to the error in these solutions if they were recalculated using 16 intervals of equal width? You must justify your answer with a reason.

(5 marks)

Solution:

We would expect that error to decrease be a factor of 16.

3 marks

This is because we have approximated the derivatives using second-order finite-difference approximations and since the new step length is $\frac{1}{4}$ of the initial step length then the error will decrease by $(\frac{1}{4})^2 = \frac{1}{16}$.

2 marks

6. (a) Simpson's rule for computing a definite integral is

$$\int_{x_0}^{x_n} f(x)dx = \frac{h}{3} \left(f(x_0) + f(x_n) + 4 \sum_{i=1,3,\dots}^{n-1} f(x_i) + 2 \sum_{i=2,4,\dots}^{n-2} f(x_i) \right),$$

where $h = \frac{x_n - x_0}{n}$. Use 6 strips to approximate the following integral to 5 decimal places

$$I = \int_0^2 e^{-x^3} dx.$$

(10 marks)

Solution:

$h = \frac{2-0}{6} = \frac{1}{3}$. Summing the odd strips

$$\begin{aligned} \sum_{i=1,3,5} f(x_i) &= f(x_1) + f(x_3) + f(x_5) = f\left(\frac{1}{3}\right) + f(1) + f\left(\frac{5}{3}\right) \\ &= 0.963640 + 0.367879 + 0.009758 = 1.341278. \end{aligned}$$

4 marks

Summing the even strips

$$\begin{aligned} \sum_{i=2,4} f(x_i) &= f(x_2) + f(x_4) = f\left(\frac{2}{3}\right) + f\left(\frac{4}{3}\right) \\ &= 0.743567 + 0.093446 = 0.837013. \end{aligned}$$

3 marks

So the integral is

$$\begin{aligned} I &\approx \frac{1}{3} (f(0) + f(2) + 4f(\text{odd}) + 2f(\text{even})) \\ &= \frac{1}{9} (1 + 0.000335 + 4(1.341278) + 2(0.837013)) = 0.893275. \end{aligned}$$

So $I = 0.89328$ correct to 5 decimal places.

3 marks

(b) The 3-point Gaussian quadrature formula takes the form

$$\int_{-1}^1 f(t)dt = w_1 f(t_1) + w_2 f(t_2) + w_3 f(t_3),$$

where t_i can be determined by the roots of a Legendre polynomial

$$P_{m+1}(t) = \frac{1}{m+1} ((2m+1)tP_m(t) - mP_{m-1}(t)),$$

where $P_1(t) = t$. Given the weighting factors $w_1 = w_3 = 0.55556$ and $w_2 = 0.88889$, use the 3-point Gaussian quadrature formula to approximate

$$\int_1^4 \frac{x}{\sqrt{2+x}} dx,$$

correct to 4 decimal places.

(10 marks)

Solution:

The third-order Legendre polynomial is

$$\begin{aligned}P_3(t) &= \frac{1}{3}(5tP_2(t) - 2P_1(t)) \\&= \frac{1}{3}\left(5t\left(\frac{1}{2}(3t^2 - 1)\right) - 2t\right) \\&= \frac{t}{3}\left(\frac{5}{2}(3t^2 - 1) - 2\right) \\&= \frac{t}{3}\left(\frac{15}{2}t^2 - \frac{9}{2}\right)\end{aligned}$$

The quadrature points are the roots of $P_3(t)$ so

$$\begin{aligned}0 &= \frac{t}{3}\left(\frac{15}{2}t^2 - \frac{9}{2}\right) \\0 &= \frac{t}{3} \quad \therefore t = 0, \\0 &= \frac{15}{2}t^2 - \frac{9}{2} \quad \therefore t = \pm\sqrt{\frac{3}{5}} \approx 0.77460\end{aligned}$$

Therefore $t_1 = -0.77460$, $t_2 = 0$ and $t_3 = 0.77460$ correct to 5 decimal places.

3 marks

$$\int_1^4 \frac{x}{\sqrt{2+x}} dx$$

Let $x = at + b$ then

$$\begin{array}{lll}t = -1, & x = 1 & \therefore 1 = -a + b, \\t = 1, & x = 4 & \therefore 4 = a + b,\end{array}$$

so $a = \frac{3}{2}$, $b = \frac{5}{2}$ and $x = \frac{3}{2}t + \frac{5}{2}$.

2 marks

$$\frac{dx}{dt} = \frac{3}{2} \quad \therefore dx = \frac{3}{2}dt$$

so

$$\int_{-2}^4 \frac{x}{\sqrt{2+x}} dx = \frac{3}{2} \int_{-1}^1 \frac{\frac{3}{2}t + \frac{5}{2}}{\sqrt{\frac{3}{2}t + \frac{9}{2}}} dt$$

Evaluate the function at t_1 , t_2 and t_3

$$f(-0.77460) = \frac{\frac{3}{2}(-0.77460) + \frac{5}{2}}{\sqrt{\frac{3}{2}(-0.77460) + \frac{9}{2}}} = 0.73239,$$

$$f(0) = \frac{\frac{3}{2}(0) + \frac{5}{2}}{\sqrt{\frac{3}{2}(0) + \frac{9}{2}}} = 1.17851,$$

$$f(0.77460) = \frac{\frac{3}{2}(0.77460) + \frac{5}{2}}{\sqrt{\frac{3}{2}(0.77460) + \frac{9}{2}}} = 1.53895.$$

3 marks

Therefore,

$$\int_1^4 \frac{x}{\sqrt{2+x}} dx \approx \frac{3}{2} (0.55556(0.73239) + 0.88889(1.17851) + 0.55556(1.53895))$$

$$= 3.46415.$$

So the value of the definite integral is $I = 3.4642$ correct to 4 decimal places.

2 marks

- (c) Give ONE advantage and ONE disadvantage of using Gaussian quadrature over other numerical integration methods.

(5 marks)

Solution:

Advantage: higher accuracy using the same number of function evaluations

Disadvantage: less straight forward to implement and requires mapping of integral range to $[-1, 1]$

5 marks